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An Entity-Relation Extraction Method Based on the Mixture-of-Experts Model and Dependency Parsing

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Abstract: Entity-relation extraction (ERE) aims to identify entity types and the relationships between them from unstructured texts and is one of the key technologies for constructing knowledge graphs. However, ERE tasks face challenges such as insufficient semantic representations and the complexity of relationship types, which lead to the difficulty of triplet extraction. To address these issues, we propose an entity-relation extraction model that incorporates dependency parsing and a mixture-of-experts architecture. Specifically, we use BERT as a character encoder, while integrating dependency syntax information as a separate encoding path. We apply additive attention to fuse the two pathways of encoding, assigning different weights to each vector in the encoding layer output through a learned weighting process. This enables the model to flexibly adjust the attention given to different features, allowing for a more accurate identification and utilization of syntactic dependencies within a sentence. In the relation classification layer, we employ a mixture-of-experts architecture, allowing each expert to focus on learning different relationship labels, thereby enhancing the model's ability to accurately identify and capture specific entity relationships. The proposed model achieves superior results to the baseline models on two public ERE datasets, providing a novel and effective solution for entity-relation extraction tasks.

Keywords: entity-relation extraction; dependency parsing; additive attention; mixture-of-experts

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1. Introduction

The goal of entity-relation extraction is to simultaneously identify entity pairs and their corresponding relationships from unstructured texts. It is a core task in knowledge extraction and is crucial for downstream tasks, such as knowledge graph construction and question-answering systems. Therefore, exploring effective methods for entity-relation extraction from unstructured texts is of significant importance.

The entity-relation extraction task consists of two key subtasks: named-entity recognition (NER) [1] and relation extraction (RE) [2]. The former identifies various entities within sentences, while the latter extracts the relationship between any two entities in a sentence. The development of entity-relation extraction techniques mirrors the history of artificial intelligence, having undergone three iterations: rule-based and dictionary-based methods, statistical machine learning-based methods, and deep learning-based methods

[3]. Entity-relation extraction methods based on deep learning can be broadly categorized into two approaches: the pipeline extraction method and the joint extraction method. The pipeline approach treats NER and RE as two independent tasks, which leads to a lack of interaction between the subtasks. As a result, textual information is not fully utilized, causing issues such as redundant entities that can affect the quality of subsequent relation extraction results. In recent years, joint extractions have garnered increasing attention from researchers, which unify the two subtasks of NER and RE using a joint framework to extract relation triplets. Although joint methods have demonstrated promising performance, several challenges persist. First, there remains the issue of insufficient semantic representation. Pre-trained language models, such as BERT, are often used as encoders due to their excellent performance. However, these models cannot explicitly model syntactic structures and are limited in their ability to fully capture semantic information, which restricts their accuracy in understanding complex linguistic phenomena. Some studies incorporate syntactic information as additional features into the model but typically treat dependency syntactic information and semantic representations as separate steps or only concatenate them without fully leveraging the syntactic structure to enhance semantic understanding. Second, a text often contains complex relationship types, and the distribution of different relationship types is imbalanced, which hinders the model's ability to comprehensively understand the relationships, making triplet extraction more challenging. Figure 1 illustrates the distribution of relationship frequencies in the training set of the Chinese public dataset DUIE, which contains 48 types of relationships, with a highly skewed distribution of instances across these relationships. This is a common phenomenon in textual data. Addressing these challenges can enhance the accuracy of entity-relation extraction tasks, which in turn has a significant impact on related fields. For instance, knowledge graphs build relationship maps between entities, providing rich structured data for information retrieval, reasoning, and decision-making. High-quality entity-relation extraction ensures the accuracy and completeness of knowledge graphs, thereby improving the capability of knowledge reasoning. In question-answering systems, accurately understanding the entities and their interrelationships in a query is crucial for providing precise answers and improving the reasoning abilities for complex questions.

Therefore, to address these challenges, we propose a novel joint entity and relation extraction model. For challenge 1, we incorporated dependency syntax information at the encoding layer, encoding it separately in a single path, and applied an additive attention mechanism to fuse word-level information with BERT's character-level representations. This enables the model to recognize syntactic dependencies between words, enhancing its semantic understanding. For challenge 2, we designed multiple experts to handle different types of entity pairs and relationship combinations, thereby improving the triplet extraction capability.

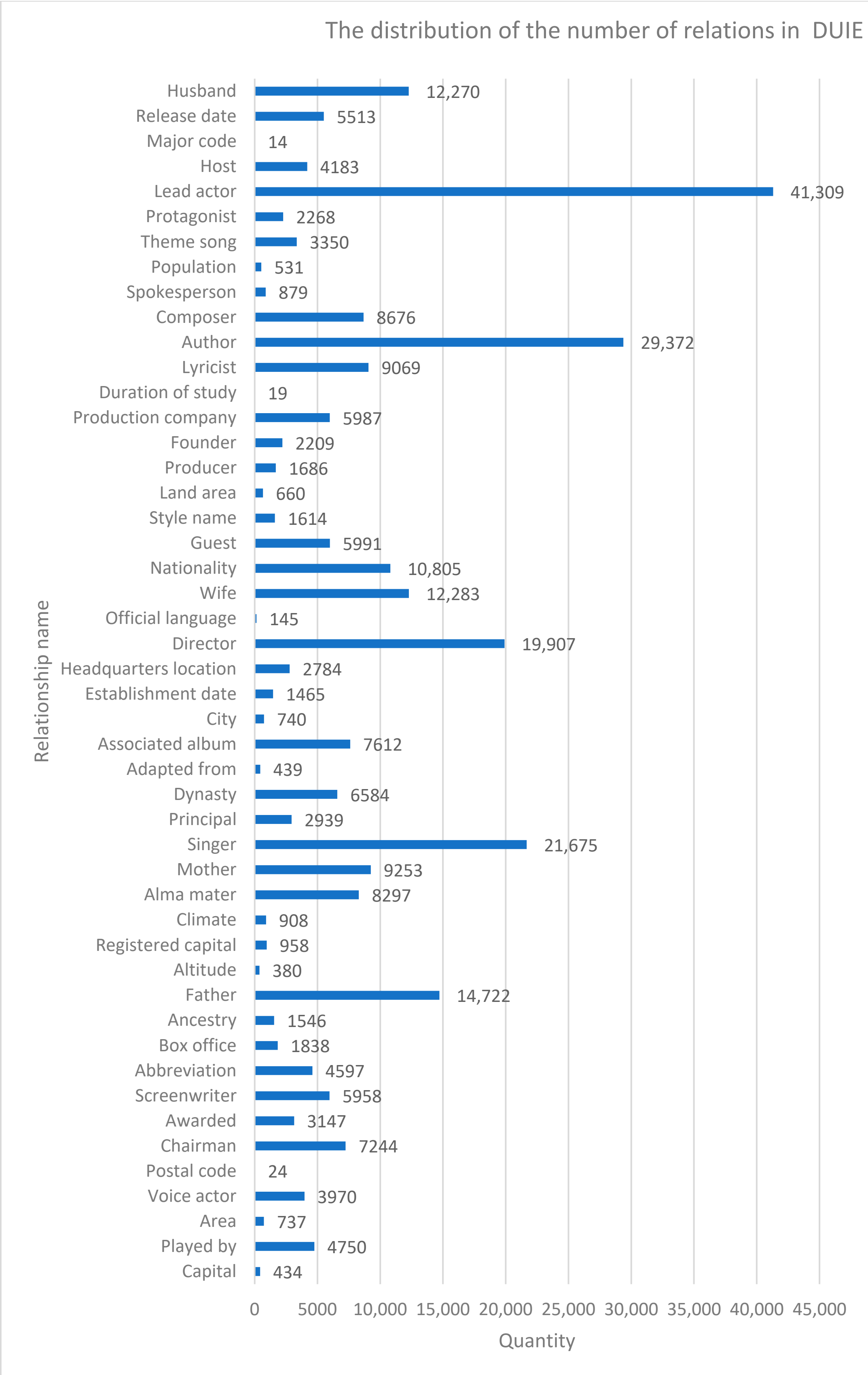

Figure 1. The distribution of the number of relations in DUIE.

The primary contributions of this paper are as follows:

1. We encoded syntactic information in a separate path, embedded it at the word level, and fused it with BERT outputs using an additive attention mechanism, allowing the

model to learn richer feature representations and to better understand the relationships between entities.
2. We introduced a mixture-of-experts architecture in the relation classification layer, where a gating network assigns weights to each expert, enabling more accurate handling of various relation extraction tasks.
3. We conducted experimental evaluations on a Chinese public dataset and an English public dataset. The results demonstrate that the proposed method outperforms other baseline models in terms of performance.

The rest of this paper is structured as follows: Section 2 provides an overview of related work. Section 3 describes the architecture of the proposed model. Section 4 details the datasets and experimental setup, followed by an analysis of the experimental results. Section 5 presents the conclusions and directions for future work.

2. Literature Review

2.1. Recent Advances in Entity-Relation Extraction

2.1.1. Pipeline Methods

Current entity-relation extraction methods can be divided into two main categories: pipeline methods and joint extraction methods. In pipeline methods, researchers first build an entity recognition model using manually designed feature extraction and kernel function methods and then construct a model that can recognize semantic relations based on entity pairs to achieve relation extraction between entities [4]. With the rapid development of deep learning techniques in recent years, some end-to-end deep learning models have gradually dominated this field. Research related to deep learning-based named-entity recognition (NER) and relation extraction (RE) has achieved substantial progress [5–8]. Li et al. [9] modeled the NER problem by leveraging relationships between words, proposing a new data-encoding pattern and a novel model architecture, which performed excellently on nested, non-nested, and discontinuous datasets. Qi et al. [10] proposed a CNER model based on text similarity and mutual information maximization, where potential word boundaries are identified by calculating text similarity, and mutual information between characters, potential word boundaries, and sentences is maximized to enhance input features. Additionally, a specific network architecture was designed for structural enhancement. Zhao et al. [11] enhanced dialog-based relation extraction by predicting the speaker and trigger words. Speaker prediction captures features of the speaker who presents relevant entities, while trigger word prediction captures contextual information between entity relations, thus enabling relation extraction.

2.1.2. Joint Extraction Methods

Since the annotated data used in relation extraction tasks originate from the previously named-entity recognition task, there is a certain correlation between the two tasks. However, in pipeline methods, the named-entity recognition task and the relation extraction task are learned and trained independently, which leads to a lack of interaction between the two subtasks, weakening the connection between them, causing underutilization of textual information, and resulting in issues like redundant entities. These issues lead to errors that, in turn, affect the quality of relation extraction results. Therefore, joint extraction methods have become mainstream in recent years, aiming to use the named-entity recognition task to better assist entity-relation extraction. Miwa and Bansal [12] first employed a neural network-based approach for joint extraction, capturing both word-sequence and dependency-tree-substructure information by stacking a bidirectional tree structure on a bidirectional LSTM-RNN. In this model, the sequence-labeling task and the prediction task are carried out simultaneously, and the predicted results are obtained through a decoder, with parameter updates performed during the training process using the backpropagation algorithm. Ding et al. [13] proposed an architecture for joint entity and relation extraction, using external lexical and syntactic knowledge to address the limitations encountered by BERT-based models in joint extraction tasks within specific Chinese domains. Wei et al. [14] proposed a cascade binary-labeling framework named CASREL for relation triplet extraction. The model first obtains the boundary information of the head entity via sequence labeling, and then, for each relation, a binary sequence labeler identifies the tail entity corresponding to the head entity in the word–relation pair, ultimately achieving joint entity-relation extraction. Shang et al. [15] developed a fine-grained triplet classification model, OneRel, which follows a single-module, single-step modeling approach, directly extracting triplets from text statements, achieving performance superior to previous studies on public datasets. Lu et al. [16] proposed a Universal Information Extraction (UIE) framework, which unifies the extraction of different structures by encoding structured extraction information and then adaptively generates target extractions based on a pattern extraction mechanism and a structure-guided controller. The aforementioned methods primarily utilize BERT as the pre-trained model, which is a typical representative of transformer-based approaches. However, pre-trained language models like BERT do not fully capture semantic information and often neglect the modeling of syntactic structures. This limitation hinders the model's ability to accurately interpret complex linguistic phenomena.

In recent years, there has been a growing amount of content on social media that is represented jointly in multimodal forms, such as texts, images, and audio. Multimodal entity-relation extraction models have become an important emerging research direction in the field of natural language processing. Wan et al. [17] proposed a multimodal social relation extraction method based on few-shot learning, which uses limited data to learn and extract social relations from both texts and facial images. This marks the first attempt to incorporate both texts and images for social relation extraction. Hu et al. [18] introduced an entity–object and relation–image alignment pre-training task, which extracts self-supervised signals from large volumes of unlabeled image–caption pairs to pre-train the multimodal fusion module and enhance MERE performance. However, multimodal entity-relation extraction models typically require large amounts of labeled data across various types, and for traditional domains that rely solely on text data, the benefits of multimodal methods are not as apparent. As a result, the range of applicable scenarios is somewhat limited.

2.2. Dependency Parsing in NLP

Dependency parsing (DEP) is a crucial natural language processing task [19] aimed at identifying syntactic relationships between words in a sentence. It has been widely applied in sentiment analyses [20], opinion role labeling (ORL) [21], and other fields. Kambhatla [22] was the first to apply dependency parsing to the entity-relation extraction field, demonstrating that introducing dependency syntax information and performing multi-feature fusion can enrich sentence feature representations and improve model performance in entity-relation extraction tasks. The TaMM method proposed by Chen et al. [23] maps words and dependency relations that are directly linked to entities in the syntactic dependency tree to memory slots and assigns a weight to each slot based on its contribution to relation extraction. Experiments showed that the incorporation of specific dependency relations effectively enhanced triplet extraction performance. The attention-guided graph convolutional network proposed by Guo et al. [24] uses the full dependency tree as the input to the graph convolutional network, and during training, it learns to selectively focus on dependency substructures, ultimately achieving superior performance across multiple relation extraction tasks. Tian et al. [25] employed an attention-based graph convolutional neural network model to identify noise in syntactic knowledge using pruning-based dependency syntax, thereby enhancing model performance in relation extraction tasks. In summary, dependency parsing methods can effectively improve the performance of joint entity-relation extraction. However, current research typically treats dependency syntactic information and semantic representations as separate steps or incorporates syntactic information as additional features into the model, lacking effective integration of syntactic structure and semantic understanding. Our model incorporates dependency syntax information into the encoding layer and employs an additive attention mechanism to assign varying weights to the vectors' output by the encoding layer. This enables a more flexible and effective integration of syntactic and semantic information, thereby addressing the issue of insufficient syntactic information extraction in pre-trained language models.

2.3. Applications of MoE in NLP

The concept of mixture-of-experts (MoE) was first introduced in the literature [26]. Due to its flexibility, scalability, low computational cost, strong specialization, and generalization capabilities, MoE has been widely used in AI subfields, such as computer vision [27], multimodal learning [28], speech recognition [29], and recommendation systems [30], achieving great success. In the natural language processing field, research on MoE models mainly focuses on reducing the computational cost of large language models by decomposing large models into smaller expert components, thereby improving computational efficiency and optimizing resource allocation [31]. MoE research has become one of the effective solutions to reduce the computational costs of large models. Drawing inspiration from the MOE principle and its applications, and considering the significant imbalance in the sample sizes of different relation types in entity-relation extraction tasks, we innovatively introduce a mixed expert architecture into the relation classification layer of our model. By using a gating network, tasks are assigned to each expert based on their specialization, allowing the model to better address the data-imbalance issue and enhance the accuracy of triplet extraction.

3. Methodology

The entity-relation extraction model proposed in this paper consists of four main components: an encoding layer, a feature fusion layer, a relation classification layer, and a decoding layer. The detailed architecture of the model is illustrated in Figure 2.

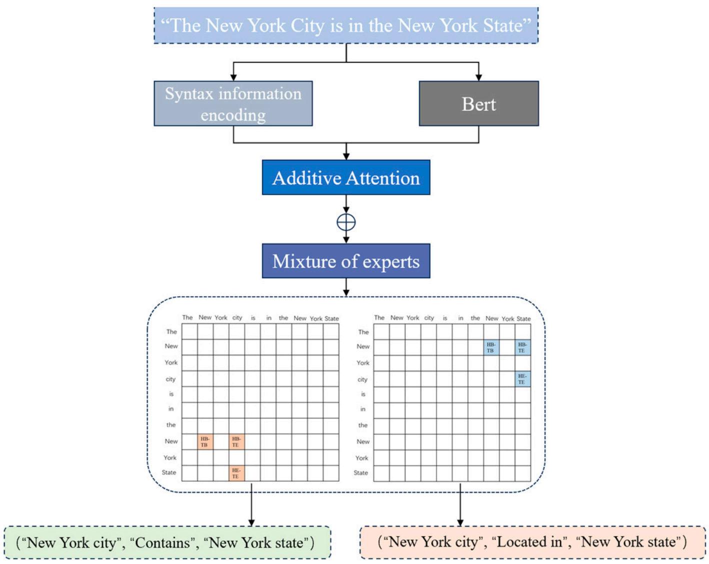

Figure 2. The overall architecture of the proposed model.

In the encoding layer, the BERT model is initially employed to encode the input text sequence and generate vector representations for each character. To compensate for BERT's limitations in capturing syntactic information, we introduce an additional dependency syntax encoding module, as detailed in Section 3.1.2. The feature fusion layer integrates character-level and word-level information, enabling the model to learn more comprehensive feature representations and understand the relationships between entities in a sentence more accurately, as explained in Section 3.2. Notably, we innovatively introduce the concept of a mixture-of-experts architecture in the relation classification layer, incorporating a gating network to dynamically select the appropriate expert based on the input sample's features, as discussed in Section 3.3. The decoding layer adopts the approach of the OneRel [15] model, utilizing a specific relation-tagging strategy to define the boundaries of the head and tail entities and decode entities and relations from the output matrix. This method effectively minimizes redundant information and prevents cascading errors.

3.1. Encoding Layer

3.1.1. BERT Encoding

The proposed model employs BERT as the base encoder. The input to BERT consists of three components: word embeddings, sentence embeddings, and position embeddings. Word embeddings are used to represent each word or subword in a sentence, while sentence embeddings distinguish between different sentences, and position embeddings capture the relative positions of words within a sentence. For a sentence $S = \{t_1, t_2, \ldots, t_n\}$, composed of n characters, it is processed by the pre-trained BERT model to obtain the corresponding representations for each character.

$$\{b_1, b_2, \ldots, b_n\} = BERT(\{t_1, t_2, \ldots, t_n\})$$

In this context, b_i denotes the semantic representation of character t_i, produced by BERT.

3.1.2. Syntax-Information Encoding

Since the output of BERT is based on character-level granularity, it cannot fully capture the syntactic structure of entities and their relationships. To address this, we introduce an additional syntax-information encoding module, which enhances the model's ability to capture the syntactic relationships between entities by effectively modeling word-level syntactic features. Initially, the input text is tokenized into words, yielding a set of word-level representations, $V = \{v_1, v_2, \ldots, v_n\}$, where n represents the total number of words. These words are then mapped onto a higher-dimensional vector space via an embedding layer to obtain word-level embeddings:

$$\{e_1, e_2, \ldots, e_n\} = Embedding(\{v_1, v_2, \ldots, v_n\})$$

where e_i represents the initial embedding vector of word v_i. To capture word-level contextual information, BiLSTM is used to learn the contextual relationships at the word level. BiLSTM effectively models the sequential order between words and bidirectional dependencies, outputting the word's contextual representations:

$$\{m_1, m_2, \ldots, m_n\} = BiLSTM(\{e_1, e_2, \ldots, e_n\})$$

where m_i represents the word v_i's contextual semantic feature, incorporating both its preceding and succeeding contextual information. For the given input dependency parsing, we construct connections between the source and target nodes for each word pair with a dependency relationship, resulting in a dependency syntax graph. We then model the word-to-word relationships in the dependency syntax graph using GNN and use GAT (Graph Attention Network) to learn the propagation and updating of dependency relationships in the sentence layer by layer. Each word is represented as a node in the dependency syntax graph, with edges representing the syntactic relationships between words. GNN captures syntactic information based on the graph structure to obtain a deep representation of each word:

$$\{g_1, g_2, \ldots, g_n\} = GNN(\{m_1, m_2, \ldots, m_n\})$$

where g_i denotes the syntax-enhanced feature after the GNN processes the dependency syntax graph. During the process of updating the information between nodes in the graph, each node aggregates information from its neighboring nodes while integrating its features for updates. This enables the model to capture long-range dependencies between words and effectively incorporate this information into the representation of each word. Specifically, when GNN propagates information between nodes, it not only considers the contextual semantics of the words but also incorporates structural information derived from dependency syntax, thus more accurately capturing the structured dependency relationships between words.

3.2. Feature Fusion Layer

To integrate word-level information with the character-level information from BERT, inspired by the work in [32], we introduce an additive attention mechanism. This mechanism learns weighted combinations, assigning different weights to each vector output by the encoding layer, enabling the model to perform information fusion more flexibly and reduce the interference of redundant information. The architecture of the additive attention mechanism is shown in Figure 3.

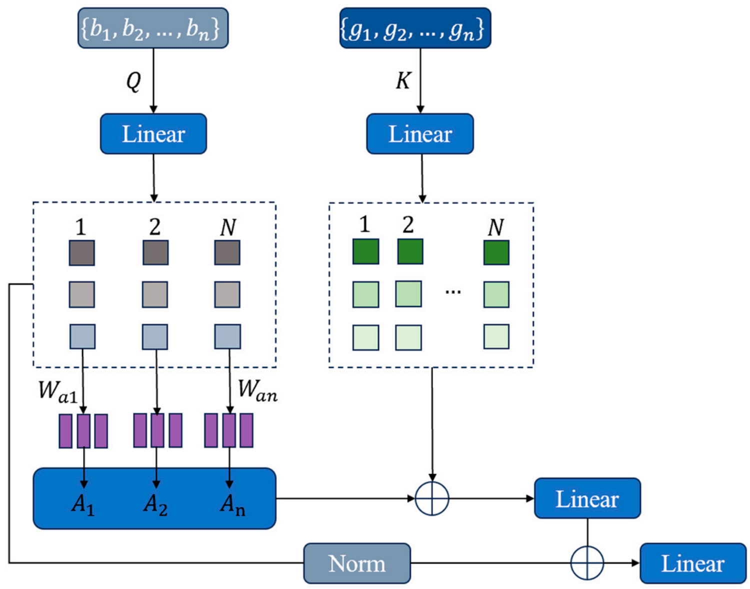

Figure 3. Additive attention mechanism.

First, two different linear transformation matrices are used to generate the corresponding query (Q) and key (K) matrices, respectively, for the BERT output and the GNN output:

$$Q = Linear_Q(BERT)$$

$$K = Linear_K(GNN)$$

For each BERT input sequence, we multiply the query matrix Q by a learnable parameter vector W_a to calculate the attention weight for each token:

$$A_{weight} = Q \cdot W_a$$

The attention weight represents the relevance of each token in the input sequence to the global context. Here, A_{weight} is the attention score for each token. Next, we normalize and scale the attention scores to obtain the attention weights A:

$$A = Softmax(\frac{A_{weight}}{\sqrt{d}})$$

where d is the dimension of the query vector. The scaling operation helps stabilize gradients and prevent overlarge numerical values. The Softmax operation ensures that the sum of the attention weights for all tokens equals 1, allowing the weight distribution to reflect the importance of each token. By weighting and summing the query matrix with the attention weights A, we obtain the global attention vector q_{global}:

$$q_{global} = \sum_{i=1}^{n} A_i \cdot Q_i$$

Here, q_{global} is the global query vector, representing the weighted aggregation of information, which captures the most important features in the input sequence. Next, the global query vector q_{global} undergoes an element-wise product with the matrix K to obtain the global context representation:

$$Context = q_{global} \odot K$$

Finally, a linear transformation layer is applied to further process the global context representation, yielding the final hidden representation h_i for each token:

$$h_i = Linear_{final}(Context_i) + Norm(Q)$$

where h_i is the final token representation, incorporating both the semantic information from BERT and the syntactic information from the GNN. This enables the model to better understand the context and the dependency relationships between words. Norm refers to the normalization operation.

3.3. Relation Classification Layer

The classification of the triplet's category can be achieved by enumerating all possible combinations of (hi, rk, hj) and designing a classifier to assign the corresponding high-confidence labels, where rk is the randomly initialized relation representation, and hi and hj are, respectively, the representations of the head and tail entities. In this layer, different experts are implemented using MLPs with different weights. The architecture of the mixture-of-experts model is shown in Figure 4. The gating network receives the data and selects the most suitable expert model for processing based on the data's features, calculating the corresponding scores for the relation. Let G denote the gating network, with the input data being x; the output of the gating network is the weight distribution across the expert models:

$$G(x) = [w_1, w_2, \dots w_n]$$

where w_i represents the weight of the i-th expert model, and $\sum w_i = 1$.

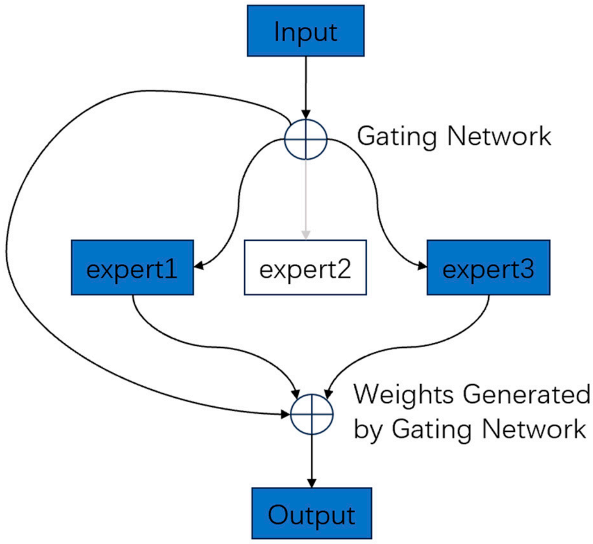

Figure 4. Mixture-of-experts model architecture.

Let the *i*-th expert model be denoted as E_i, and the triplet score matrix computed by E_i for the input data x is:

$$S_i = E_i(x)$$

where S_i is a two-dimensional matrix, and the elements in the matrix represent the scores for the head entity, tail entity, and the possible relations between them. The final triplet score matrix S is the weighted combination of the triplet score matrices calculated by each expert model:

$$S = \sum_1^n (w_i * S_i)$$

3.4. Decoding Layer

To decode entities and relationships from the output matrix accurately and effectively, a specific relation-based indexing strategy is employed to determine the boundary markers of the head and tail entities. For a given sentence, a classifier is designed to assign labels to all possible combinations. A three-dimensional matrix is maintained to store the classification results. An entity can be identified by detecting its boundary markers, and four types of labels are used in the labeling strategy.

(1) HB-TB: The head entity's starting token is connected with the tail entity's starting token. For example, there is a relationship between the entities "New York City" and "New York State", so the combined classification label ("New", "Located in", "New") is assigned as "HB-TB."

(2) HB-TE: The head entity's starting token is connected with the tail entity's ending token. For instance, "New" is the starting token of "New York City", and "State" is the ending token of "New York State", so the combination ("New", "Located in", "State") is designated as "HB-TE."

(3) HE-TE: The head entity's ending token is connected with the tail entity's ending token. For example, the combination ("City", "Located in", "State") is designated as "HE-TE."

(4) "-": No connecting relationship exists.

Using the triangular labeling information from the matrix, the entity pairs for each relation can be determined. For each relation, the span of the head entity extends from "HB-TE" to "HE-TE"; the span of the tail entity extends from "HB-TB" to "HB-TE." The two paired entities share the same "HB-TE", thus satisfying the conditions for the relation triplet (s, r, o).

4. Results

4.1. Datasets

To evaluate the effectiveness of the proposed model, we conducted experiments on a Chinese public dataset and an English public dataset. Table 1 provides the details of each dataset. The DUIE dataset includes 48 distinct relationship types, with the corpus sourced from Baidu Baike, Baidu news feeds, and Baidu Tieba texts. It comprehensively covers both formal and informal expressions, allowing for a thorough evaluation of relationship extraction capabilities in real-world business scenarios. The NYT dataset, constructed through distant supervision, comprises 24 predefined relationship types.

Table 1. Statistics of the datasets.

Datasets	Training Set	Validation Set	Test Set	Relation Type
DUIE	169,744	20,497	20,500	48
NYT	56,195	5000	5000	24

4.2. Experiment Setup

The code for the proposed model is implemented using the Pytorch 2.3.1 framework. The Chinese pre-trained model is based on the bert-base-Chinese model released by Google, while the English pre-trained model utilizes the bert-base-cased model. To prevent overfitting, a dropout rate of 0.3 is applied. The initial learning rate is set to 2×10^{-5}, and the batch size is set to 4. The proposed model, along with several baseline models, is trained and tested using an NVIDIA A40 GPU.

Three commonly used evaluation metrics are employed to evaluate the model's performance: precision, recall, and F1 score. Precision represents the proportion of true entities among all predicted entities, recall represents the proportion of true entities correctly predicted by the model out of all true entities, and the F1 score is the harmonic mean of precision and recall. The formulas for these metrics are provided below:

$$Precision = \frac{\sum_{i=1}^{c} TP_i}{\sum_{i=1}^{c} TP_i + \sum_{i=1}^{c} FP_i}$$

$$Recall = \frac{\sum_{i=1}^{c} TP_i}{\sum_{i=1}^{c} TP_i + \sum_{i=1}^{c} FN_i}$$

$$F1 = 2 * \frac{Precison * Recall}{Precision + Recall}$$

In addition, to evaluate the model's reliability, we use bootstrapping to compute the confidence intervals for accuracy, recall, and F1 score, with a 95% confidence level. We also employ the paired t-test to assess whether the performance difference between the proposed model and the baseline model is statistically significant, with the significance level set at $p < 0.05$.

4.3. Comparison with Basic Models

To evaluate the superiority of our proposed entity-relation joint extraction model, we conducted comparative experiments with CasRel [14], TPLinker [33], CasDE [34], and OneRel [15] on the DUIE and NYT datasets.

(a) CasRel: a model that utilizes BERT as the encoder and employs a multi-module, multi-step approach for relation triplet extraction.

(b) TPLinker: a model that introduces a handshake annotation method that enables the extraction of triplets in a single decoding pass, while addressing issues such as overlapping relations and exposure bias.
(c) CasDE: this method leverages a text-specific relation decoder to detect relations within sentences and uses a span-based labeling scheme to identify the corresponding head and tail entities.
(d) OneRel: a single-step, single-module joint extraction strategy that reformulates the joint extraction task into a fine-grained triplet classification problem.

The experimental results are shown in Tables 2 and 3, which indicate that our model consistently outperforms various baseline models on both datasets in terms of accuracy and F1 score. Specifically, on the DUIE dataset, our model achieves remarkable results, with an accuracy of 76.87%, a recall of 76.40%, and an F1 score of 76.63%, improving by 6.06%, 1.85%, and 4.00%, respectively, compared to the backbone model OneRel. On the NYT dataset, the model achieves an accuracy of 93.6%, a recall of 92.1%, and an F1 score of 92.8%, with improvements of 1.4%, 1.5%, and 1.4%, respectively, over the backbone model OneRel. We calculated the confidence intervals for the precision, recall, and F1 score of the proposed model using bootstrapping. The results show that our model has relatively narrow confidence intervals, indicating stable performance. The validation loss curves for the proposed approach are shown in Figures 5 and 6. The curves illustrate how the model's performance evolves during training. As the number of iterations increases, the validation loss steadily decreases, indicating that the model is converging towards an optimal state. The stability of the validation loss in the later stages further demonstrates the model's effective generalization and ability to prevent overfitting.

Table 2. Experimental results of different models on DUIE.

Model	P/%	R/%	F1/%
CasRel	72.41	70.43	71.41
TPLinker	62.98	73.85	67.98
CasDE	73.39	72.26	72.82
OneRel	70.81	74.55	72.63
Ours	**76.87** ([75.25, 76.74])	**76.40** ([74.98, 76.13])	**76.63** ([75.05, 76.49])

Table 3. Experimental results of different models on NYT.

Model	P/%	R/%	F1/%
CasRel [14]	89.7	89.5	89.6
TPLinker [30]	91.4	**92.6**	92.0
CasDE [31]	89.9	91.4	90.6
OneRel	92.2	90.6	91.4
Ours	**93.6** ([92.7, 93.5])	92.1 ([91.3, 91.8])	**92.8** ([91.9, 92.3])

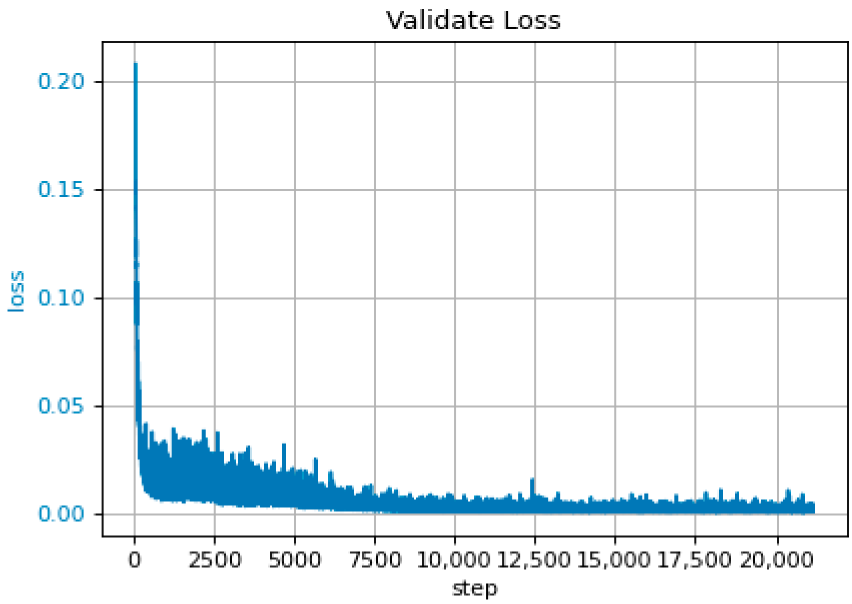

Figure 5. The loss curve of DUIE dataset on the proposed methodology.

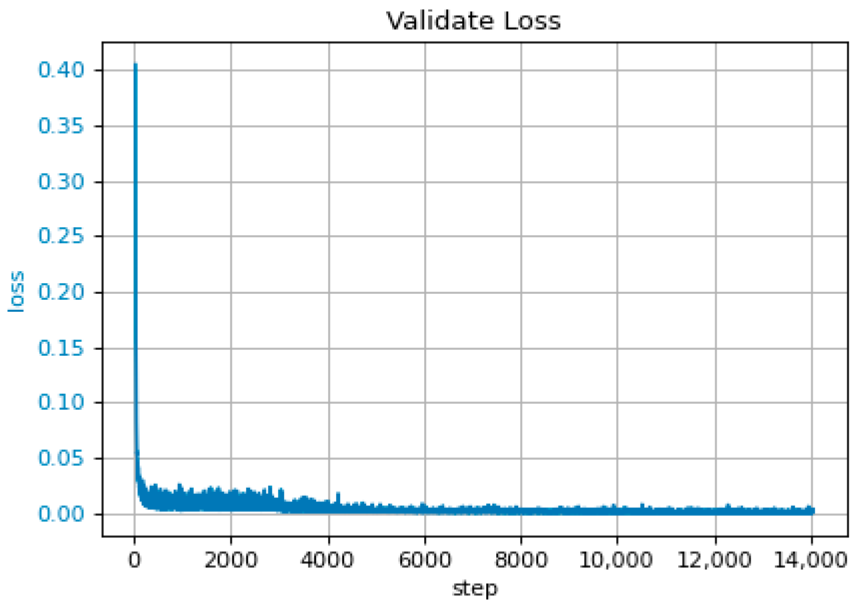

Figure 6. The loss curve of NYT dataset on the proposed methodology.

To further verify whether the performance differences between the models are statistically significant, we used the paired t-test to analyze the experimental results. When the p-value is smaller than the significance level (typically 0.05), the null hypothesis can be rejected, indicating a significant difference between the samples. On the DUIE dataset, based on the F1 scores between our model and OneRel, the paired t-test results show that the performance difference is significant (T-statistic = 5.56; p-value = 0.005). Similarly, on the NYT dataset, we conducted a similar test, and the results show that the F1 score difference between our model and OneRel is also significant (T-statistic = 4.24; p-value = 0.013). These statistical analyses further confirm the superiority of our model on both datasets.

The performance improvement of our model is more prominent on the DUIE dataset. This is due to not only the large scale of the DUIE dataset but also the higher linguistic complexity of the extraction tasks compared to English datasets as a Chinese dataset. The model proposed in this study significantly improves the ability to model complex relationships in Chinese data by incorporating dependency-graph-structure information and deeply integrating it with contextual semantic features. Additionally, the DUIE dataset contains a greater variety of relationship types, and the mixture-of-experts architecture is able to select the most suitable expert model for recognizing complex relationship types, thereby capturing and identifying specific entity relationships more accurately.

4.4. Ablation Experiment

To further evaluate the effectiveness of our proposed model in improving entity-relation extraction performance, ablation experiments were conducted on the DUIE and NYT datasets. The specific results are shown in Tables 4 and 5, where "ours-l" denotes the removal of the dependency syntax analysis module, and "ours-m" denotes the removal of the mixture-of-experts architecture.

Table 4. Ablation experiment results of the model on DUIE.

Model	P/%	R/%	F1/%
OneRel	70.81	74.55	72.63
Ours-l	75.15	76.26	75.70
Ours-m	75.65	75.53	75.59
Ours	**76.87**	**76.40**	**76.63**

Table 5. Ablation experiment results of the model on NYT.

Model	P/%	R/%	F1/%
OneRel	92.25	90.62	91.42
Ours-l	93.33	90.68	92.00
Ours-m	92.74	**92.54**	92.64
Ours	**93.62**	92.05	**92.83**

By comparing the ablation experiment results, it is observed that removing either the dependency syntax analysis module or the mixture-of-experts architecture leads to a decline in model performance. Specifically, after removing the syntactic information module, the F1 scores of the model decreased by 2.30% and 0.83%, respectively, on the DUIE and NYT datasets. After removing the mixture-of-experts architecture, the F1 scores, respectively, decreased by 1.04% and 0.19% on the DUIE and NYT datasets. The model proposed in this paper, by constructing a dependency-based graph neural network, clearly captures the dependency relationships between words in a sentence, particularly the dependency paths between entities, providing important prior information and semantic cues for the relation extraction task. Notably, compared to English texts, Chinese sentence structures are more complex, and word dependencies are often not confined to adjacent characters but may span across multiple characters or even phrases. The incorporation of the additive attention mechanism for feature fusion allows for a better capture of long-range dependencies between characters in a sentence. An imbalance in the distribution of relationship types in the dataset is a common issue. Introducing the mixture-of-experts (MoE) architecture can effectively mitigate conflicts between categories. Sentences containing relationship types with a higher number of occurrences are often more common in real-world texts and tend to include multiple triples, increasing the difficulty of correctly identifying all triples. On the other hand, relationship types with fewer occurrences suffer from limited training data, which leads to insufficient understanding of these relationships by the model and poorer performance in extracting the corresponding triples.

The mixture-of-experts architecture dynamically generates allocation weights for the input data across different experts through a gating network, enabling each expert to effectively model the features of different relationship types. This results in a more accurate capture of specific semantic information and entity relationships. Especially in scenarios with high label diversity or an uneven relation distribution, this architecture can dynamically adjust the learning weights for different relations, effectively alleviating conflicts among categories. The experimental results confirm that both the syntactic information module and the hybrid expert architecture significantly improve the model's accuracy.

4.5. Discussion on Computational Cost

We conducted a statistical analysis of the computational cost for the backbone model OneRel and the proposed model. The FLOPs of OneRel are 63.80 GFLOPs, and the number of parameters is 91.18 M. The FLOPs of the proposed model are 134.62 GFLOPs, and the number of parameters is 98.27 M. Compared to the baseline model OneRel, the model proposed in this paper increased both the computational complexity and parameter count. This increase primarily arises from the introduction of dependency syntax information through the GNN and the mixture-of-experts architecture. These modules increase both computational and storage demands. The graph neural network processes dependency relations through multiple layers of graph convolutions, not only making each word's representation dependent on its contextual information but also integrating syntactic dependency structures. Furthermore, the mixture-of-experts architecture introduces multiple expert networks and gating mechanisms, requiring more parameters to dynamically select experts during the learning process. Although the computational cost increased, the accuracy and expressiveness of the model improved, enabling it to better handle complex language structures and diverse relationship types, thus offering greater practical value and applicational potential.

5. Conclusions

In this work, we proposed a novel entity-relation extraction model aimed at addressing the common challenges in current entity-relation extraction tasks, such as insufficient semantic representations and the complexity of relationship types, which make triplet extraction more difficult. Specifically, to tackle the issue of insufficient semantic information extraction by existing models' encoders, we introduced dependency syntax information, providing critical semantic insights for the entity-relation extraction task. This enables the model to more accurately recognize and leverage syntactic dependencies within a sentence, enhancing its understanding of sentence structure and learning richer feature representations. In the relation classification layer, we incorporated a mixture-of-experts architecture, where a gating network dynamically generates different assignment weights for the experts, allowing the trained experts to capture specific types of relations or features, thereby improving the overall accuracy and generalization ability of the model. Through experiments on the DUIE and NYT datasets, we confirm that compared to baseline models, the proposed model demonstrates superior performance, with both the syntax information module and the mixture-of-experts architecture contributing to increased precision of entity-relation extraction. This study provides an effective approach for entity-relation extraction tasks and offers new perspectives for other similar natural language processing tasks. However, the proposed model still faces some limitations in terms of dataset diversity and computational cost. Its performance may vary across different domains and types of datasets. We could further expand the experimental scale to validate the model's reliability and practicality in subsequent research. In addition, although the model's accuracy improved, its computational cost remains high. We will consider optimizing the model's computational efficiency, for example, by using techniques such as pruning or knowledge distillation, to reduce the computational overhead.

In future work, we will explore additional applicational scenarios for the proposed model. In the multimodal entity-relation extraction task, we aim to leverage the mixture-of-experts architecture to integrate information from diverse modalities. Each expert will specialize in processing a specific type of data, extracting richer semantic features from visual cues in images and audio characteristics like tone and speech rate, such as images and audio, to further enhance the accuracy and robustness of entity-relation extraction. Additionally, applications in low-resource environments are an important avenue for future research. Our model has shown promising results on two datasets with distinct language features, providing preliminary evidence of its cross-lingual generalization capability. By using dependency parsing to capture syntactic dependencies, the model is well-suited to handle structural differences across languages. The mixture-of-experts architecture enables the efficient selection and processing of different relational types, and its flexibility enhances the model's ability to generalize across diverse linguistic data. We could combine techniques such as prompt learning and domain adaptation to extend our model to low-resource or underrepresented languages, addressing the issue of limited annotated data while broadening its potential applications.

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